

# issues

## Issues

**Revision:** 224 **Last modified:** 2026-06-16T21:12:37Z **Description:** Active workable-item registry for Helix Cluster OS **Authority:** Constitution §11.4.93 (workable-items DB single source of truth) **Generated-by:** scripts/docs/db\_to\_md.py (DB is canonical; edit via cmd/hxc-registry, not by hand)

Total active items: **223**. Canonical source: data/hxc\_registry.db. Full per-item detail (description, closure criteria, required test types, source refs) lives in the DB and docs/research/\_ledger/\*.json.

## Foundation (anti-bluff) (7 active)

| HXC     | Type    | Pri | Status | Title                                                                                     |
|---------|---------|-----|--------|-------------------------------------------------------------------------------------------|
| HXC-932 | Task    | P0  | Queued | Wire-in vs prune decision + execution for ~167 orphaned pkg packages (STRATEGIC umbrella) |
| HXC-016 | Task    | P1  | Queued |                                                                                           |
| HXC-017 | Task    | P1  | Queued |                                                                                           |
| HXC-018 | Task    | P1  | Queued |                                                                                           |
| HXC-019 | Task    | P1  | Queued |                                                                                           |
| HXC-020 | Task    | P1  | Queued |                                                                                           |
| HXC-949 | Feature | P2  | Queued | Chaos OTel polyglot trace (Rust/Elixir/Go) + Grafana dashboards (split from HXC-1258)     |

## MVP (5 active)

| HXC      | Type    | Pri | Status | Title                                                                    |
|----------|---------|-----|--------|--------------------------------------------------------------------------|
| HXC-1100 | Feature | P1  | Queued | Implement real Bazel RBE build execution (internal/build)                |
| HXC-1140 | Task    | P1  | Queued | Publish v1.0.0-dev-mvp release across all modules with packaging         |
| HXC-1128 | Feature | P2  | Queued | Implement Health Monitor with eBPF probes and LSTM failure prediction    |
| HXC-1129 | Feature | P2  | Queued | Implement CRIU/DMTCP-based session live migration orchestration          |
| HXC-1143 | Feature | P2  | Queued | Implement distcc/icecream distributed C/C++ compilation and ccache layer |

## Phase 2 (9 active)

| HXC      | Type    | Pri | Status | Title                                                                   |
|----------|---------|-----|--------|-------------------------------------------------------------------------|
| HXC-1153 | Feature | P1  | Queued | Implement pkg/resources Linux GPU reader via /sys/class/drm (NVML-free) |
| HXC-1159 | Task    | P1  | Queued | E2E: a Linux node joins the cluster mesh and becomes schedulable        |
| HXC-906  | Docs    | P1  | Queued | PHASE_2_ROADMAP.md                                                      |
| HXC-1154 | Feature | P2  | Queued |                                                                         |

| HXC      | Type     | Pri | Status | Title                                                                                             |
|----------|----------|-----|--------|---------------------------------------------------------------------------------------------------|
|          |          |     |        | Integrate universal Vulkan compute backend (vendor-neutral, no device-specific code)              |
| HXC-1155 | Feature  | P2  | Queued | Implement console/node thermal sink reader via /sys/class/thermal                                 |
| HXC-1161 | Feature  | P2  | Queued | E2E: aggregate GPU pool runs llama.cpp inference across heterogeneous nodes                       |
| HXC-1163 | Task     | P2  | Queued | Validate cross-NAT WireGuard mesh hole-punching across real hosts                                 |
| HXC-1166 | Research | P3  | Queued | CRIU/DMTCP live process migration (research + spike)                                              |
| HXC-931  | Task     | P3  | Queued | Reconcile internal/console linux_boot.go (BootMachine) with boot_coordinator.go (BootCoordinator) |

### Phase 3 (31 active)

| HXC      | Type    | Pri | Status | Title                                                            |
|----------|---------|-----|--------|------------------------------------------------------------------|
| HXC-1168 | Feature | P0  | Queued | Implement SBC Linux node agent (Orange Pi 5 Max target, Tier T3) |
| HXC-1171 | Task    | P0  | Queued | Validate Mali-G610 Vulkan/OpenCL compute on SBC                  |
| HXC-1172 | Feature | P0  | Queued | Scaffold Android Agent APK (Kotlin + NDK + Termux)               |
| HXC-1173 | Feature | P0  | Queued | Implement Android foreground service framework with START_STICKY |
| HXC-1174 | Feature | P0  | Queued |                                                                  |

| HXC      | Type    | Pri | Status | Title                                                                      |
|----------|---------|-----|--------|----------------------------------------------------------------------------|
|          |         |     |        | Integrate Android BatteryManager monitoring                                |
| HXC-1205 | Feature | P0  | Queued | Build edge setup wizard for device onboarding                              |
| HXC-1208 | Task    | P0  | Queued | Implement battery/thermal stress testing for edge devices                  |
| HXC-1169 | Feature | P1  | Queued | Implement SBCAdapter for SBC-specific hardware monitoring                  |
| HXC-1170 | Feature | P1  | Queued | Integrate RK3588 NPU via RKNN Toolkit2 C API                               |
| HXC-1176 | Feature | P1  | Queued | Implement Android Vulkan/ NNAPI compute backend                            |
| HXC-1177 | Feature | P1  | Queued | Implement QUIC transport client for Android (Tier T6)                      |
| HXC-1179 | Feature | P1  | Queued | Scaffold iOS Agent app (Swift, Tier T7 EDGE_DONOR)                         |
| HXC-1180 | Feature | P1  | Queued | Implement iOS Metal compute integration                                    |
| HXC-1182 | Feature | P1  | Queued | Implement iOS background execution scheduling (BGAppRefresh/ BGProcessing) |
| HXC-1192 | Feature | P1  | Queued | Implement unified NPU backend with ONNX model converter                    |
| HXC-1194 | Feature | P1  | Queued | Integrate MLC LLM universal engine for mobile inference                    |

| HXC      | Type    | Pri | Status | Title                                                                                  |
|----------|---------|-----|--------|----------------------------------------------------------------------------------------|
| HXC-1200 | Feature | P1  | Queued | Implement Jetson<br>CUDA-on-ARM<br>GPU backend                                         |
| HXC-1203 | Task    | P1  | Queued | Implement<br>Armbian TV-box<br>provisioning path<br>(RK3588/<br>Amlogic)               |
| HXC-1204 | Task    | P1  | Queued | Implement ADB-<br>over-WiFi fleet<br>provisioning for<br>Android cluster               |
| HXC-1206 | Feature | P1  | Queued | Build APK/IPA<br>distribution<br>system                                                |
| HXC-1209 | Task    | P1  | Queued | Validate ARM<br>SBC per-watt<br>performance<br>>=80% of x86                            |
| HXC-926  | Task    | P1  | Queued | Wire pkg/<br>powergater<br>CanAcceptWork<br>into edge/agent<br>work-acceptance<br>path |
| HXC-1181 | Feature | P2  | Queued | Implement iOS<br>CoreML / Neural<br>Engine inference<br>engine                         |
| HXC-1183 | Feature | P2  | Queued | Implement<br>HarmonyOS<br>Agent (ArkTS)<br>with Da Vinci<br>NPU inference              |
| HXC-1217 | Feature | P2  | Queued | Implement<br>internal/gpu cgo<br>vendor backends<br>(CUDA/ROCm/<br>oneAPI/MLX)         |
| HXC-1218 | Feature | P2  | Queued | Implement<br>session live<br>migration (CRIU/<br>DMTCP)                                |
| HXC-1219 | Feature | P2  | Queued | Implement<br>WireGuard<br>UPnP/NAT-PMP<br>NAT traversal                                |
| HXC-1221 | Docs    | P2  | Queued | Document Phase<br>3 edge multi-                                                        |

| HXC      | Type    | Pri | Status | Title                                                                                                              |
|----------|---------|-----|--------|--------------------------------------------------------------------------------------------------------------------|
|          |         |     |        | platform agent architecture diagrams                                                                               |
| HXC-1622 | Feature | P2  | Queued | HXC-1167 follow-up: CGO-enabled aarch64 cross-build via zig cc / aarch64-linux-gnu-gcc                             |
| HXC-1623 | Task    | P2  | Queued | HXC-1167 follow-up: validate cross-compiled helix-agent on real Orange Pi 5 Max (aarch64) execution                |
| HXC-1624 | Feature | P2  | Queued | HXC-1158 follow-up: add positive non-console SoC-match labels (RK3588/RPi/x86 server) to internal/console detector |

## Phase 4 (44 active)

| HXC      | Type    | Pri | Status | Title                                                                               |
|----------|---------|-----|--------|-------------------------------------------------------------------------------------|
| HXC-1222 | Feature | P0  | Queued | Build K3s orchestration substrate with RuntimeClasses for firecracker/kata/runc     |
| HXC-1223 | Feature | P0  | Queued | Implement Firecracker microVM provisioner for T1-T3 with vsock and virtio-net       |
| HXC-1224 | Feature | P0  | Queued | Implement Firecracker golden snapshot create/restore pipeline (<=28ms restore)      |
| HXC-1232 | Feature | P0  | Queued | Implement unified golden snapshot + COW instant-reset across all simulator backends |
| HXC-1234 | Feature | P0  | Queued | Build Rust DST engine: single-threaded SimLoop                                      |

| HXC      | Type    | Pri | Status      | Title                                                                            |
|----------|---------|-----|-------------|----------------------------------------------------------------------------------|
|          |         |     |             | with virtual clock and seeded PRNG                                               |
| HXC-1235 | Feature | P0  | Queued      | Integrate Rust turmoil for deterministic simulated TCP/UDP networking            |
| HXC-1242 | Feature | P0  | Queued      | Build Elixir/OTP Chaos Controller GenServer with supervision-tree isolation      |
| HXC-1251 | Feature | P0  | Queued      | Build Virtual Testing Controller OTP supervision tree (one_for_all)              |
| HXC-1252 | Feature | P0  | Queued      | Implement SessionManager with 50-session cap, 2h TTL, resource-quota enforcement |
| HXC-1253 | Feature | P0  | Queued      | Implement DevicePool GenServer with tier-to-simulator dispatch and health checks |
| HXC-1256 | Feature | P0  | Queued      | Expose REST API for session CRUD, device provisioning and snapshot restore       |
| HXC-1225 | Feature | P1  | Queued      | Implement QEMU/KVM ARM64 virt provisioner for T6 RK3588-approximate SBC          |
| HXC-1226 | Feature | P1  | Queued      | Integrate Cuttlefish/CrosVM provisioner for T5 Android AOSP devices              |
| HXC-1228 | Feature | P1  | Queued      | Implement Docker + binfmt_misc protocol stubs for T7 iOS and T8 HarmonyOS        |
| HXC-1250 | Feature | P1  | Queued      | Integrate Chaos Mesh CRDs (NetworkChaos, TimeChaos, StressChaos, DNSChaos)       |
| HXC-1257 | Feature | P1  | Queued      | Build Phoenix LiveView real-time test dashboard                                  |
| HXC-1262 | Feature | P1  | Queued      | Integrate CI/CD quality gates for GitHub Actions / GitLab CI / Jenkins           |
| HXC-1263 | Feature | P1  | In progress | Build WebAssembly plugin host on Wasmtime Component Model with WIT bindings      |
| HXC-1269 | Feature | P1  | Queued      | Build cmd/helix-testd daemon as OTP controller node entrypt                      |

| HXC      | Type     | Pri | Status | Title                                                                                           |
|----------|----------|-----|--------|-------------------------------------------------------------------------------------------------|
| HXC-1273 | Feature  | P1  | Queued | Implement vsock host-guest control channel for chaos-resilient telemetry                        |
| HXC-1284 | Task     | P1  | Queued | Benchmark Firecracker 5,000+ microVMs per host with KSM and memory overcommit                   |
| HXC-1285 | Task     | P1  | Queued | Run 72-hour continuous chaos soak proving no resource leaks or degradation                      |
| HXC-1286 | Docs     | P1  | Queued | Execute Production Readiness Review with 80-item checklist ( $\geq 95\%$ complete)              |
| HXC-1287 | Task     | P1  | Queued | Run full 8-tier matrix (160 nodes) end-to-end within 45-minute CI budget                        |
| HXC-907  | Docs     | P1  | Queued | PHASE_4_ROADMAP.md                                                                              |
| HXC-1227 | Feature  | P2  | Queued | Implement T4 gaming-console protocol-level constrained x86_64 VM                                |
| HXC-1246 | Feature  | P2  | Queued | Implement 6 hardware fault injectors (NMI, mem CE/UE, PCIe AER, CPU bit-flip, thermal throttle) |
| HXC-1266 | Docs     | P2  | Queued | Provide reference plugins (Rust device-sim, Zig device-sim) compiled to wasm32-wasi             |
| HXC-1270 | Feature  | P2  | Queued | Implement Firecracker MicroVM Kubernetes operator and CRD                                       |
| HXC-1271 | Feature  | P2  | Queued | Deploy WireGuard encrypted inter-host mesh DaemonSet for multi-host clusters                    |
| HXC-1272 | Feature  | P2  | Queued | Implement distributed snapshot pool backend (MinIO) and multi-host scheduling                   |
| HXC-1274 | Feature  | P2  | Queued | Implement Mininet WAN topology harness for multi-region network-stack testing                   |
| HXC-1275 | Research | P2  | Queued | Author TLA+/PlusCal formal specs for leader election and scheduler allocation                   |
| HXC-1276 | Research | P2  | Queued |                                                                                                 |



| HXC      | Type     | Pri | Status | Title                                                                                       |
|----------|----------|-----|--------|---------------------------------------------------------------------------------------------|
|          |          |     |        | Author Jepsen black-box linearizability test for task scheduling under partitions           |
| HXC-1279 | Feature  | P2  | Queued | Add eBPF/XDP line-rate programmable network fault injection (cilium/ebpf)                   |
| HXC-1283 | Docs     | P2  | Queued | Define Corellium / physical-hardware-in-the-loop path for iOS, PS4, RK3588 GPU/NPU          |
| HXC-1288 | Task     | P2  | Queued | Define polyglot interop boundaries (Rust↔Go↔Elixir↔Wasm)                                    |
| HXC-1289 | Feature  | P2  | Queued | Integrate libcluster automatic BEAM cluster formation via Kubernetes DNS                    |
| HXC-1625 | Bug      | P2  | Queued | Replicate SELinux relabel fix (HXC-1621) to Herald/containers/pkg/crossbuild duplicate copy |
| HXC-921  | Feature  | P2  | Queued | Bazel Remote Execution (REAPI) interop for build service (optional)                         |
| HXC-1277 | Research | P3  | Queued | Integrate Shadow/Phantom for unmodified-binary deterministic network simulation             |
| HXC-1280 | Research | P3  | Queued | Validate gem5 big.LITTLE (O3+Minor) scheduler fidelity for RK3588                           |
| HXC-1281 | Research | P3  | Queued | Evaluate VirGL/Venus virtual GPU for OpenGL/Vulkan workload functional testing              |
| HXC-1282 | Research | P3  | Queued | Integrate Tart OCI-native macOS/Linux VMs for iOS/macOS CI build-and-test                   |

## Phase 5 (26 active)

| HXC      | Type    | Pri | Status | Title                                                                                    |
|----------|---------|-----|--------|------------------------------------------------------------------------------------------|
| HXC-1314 | Feature | P0  | Queued | Implement pkg/handheld Steam Deck agent (Vulkan compute, battery, thermal) (DEFERRED hw) |

| HXC      | Type    | Pri | Status | Title                                                                                                |
|----------|---------|-----|--------|------------------------------------------------------------------------------------------------------|
| HXC-1312 | Feature | P1  | Queued | Build multi-arch universal agent container (amd64/arm64/riscv64) via buildx + distroless             |
| HXC-1313 | Feature | P1  | Queued | Implement riscv64 cross-compilation pipeline + RVV extension detection (DEFERRED CI)                 |
| HXC-1315 | Feature | P1  | Queued | Package Steam Deck Flatpak agent with systemd integration (DEFERRED hw)                              |
| HXC-1316 | Feature | P1  | Queued | Implement internal/sbc: Jetson TensorRT backend ( $\geq 60$ TOPS) (DEFERRED hw)                      |
| HXC-1317 | Feature | P1  | Queued | Implement internal/sbc RK3588 NPU + Turing RK1 cluster support                                       |
| HXC-1320 | Feature | P1  | Queued | Implement internal/enterprise EPYC/Ampere auto-provisioning + Coreboot trust detection (DEFERRED hw) |
| HXC-1321 | Feature | P1  | Queued | Implement internal/iot OpenWrt router gateway agent (GL-MT6000, Docker, $< 5\%$ overhead)            |
| HXC-1322 | Feature | P1  | Queued | Implement internal/iot NAS storage-node agent (Synology/QNAP Docker, dual-role)                      |
| HXC-1327 | Feature | P1  | Queued | Implement gVisor/Kata per-tier sandbox hardening for UNTRUSTED/edge nodes (DEFERRED)                 |
| HXC-1329 | Feature | P1  | Queued | Implement hybrid cloud-on-prem WireGuard mesh                                                        |

| HXC      | Type    | Pri | Status | Title                                                                                                    |
|----------|---------|-----|--------|----------------------------------------------------------------------------------------------------------|
| HXC-1332 | Task    | P1  | Queued | joining spot workers to on-prem cluster<br>Phase 5a integration test: 10-node mixed handheld+SBC cluster |
| HXC-1318 | Feature | P2  | Queued | Implement pkg/fpga + internal/fpga hard-SoC/soft-core/DPU backends (DEFERRED hw)                         |
| HXC-1319 | Feature | P2  | Queued | Implement bitstream verification + management for FPGA tiers (T11 required)                              |
| HXC-1324 | Feature | P2  | Queued | Implement internal/exotic Groq LPU inference adapter (<100ms TTFT) (DEFERRED cloud)                      |
| HXC-1325 | Feature | P2  | Queued | Implement internal/exotic Cerebras CS-3 large-model inference adapter (DEFERRED cloud)                   |
| HXC-1333 | Task    | P2  | Queued | Phase 5b integration test: 8-node arm64+riscv64+fpga heterogeneous cluster                               |
| HXC-1617 | Task    | P2  | Queued | Live-cloud IMDS interruption integration on real AWS/Azure/GCP spot accounts                             |
| HXC-1618 | Task    | P2  | Queued | Validate pkg/benchmark normalized scores across two physically different nodes                           |
| HXC-928  | Task    | P2  | Queued | Execute linux fixtures on real linux CI for wave-74 packages (CLAUDE-2)                                  |
| HXC-929  | Task    | P2  | Queued | Real macOS thermal source for                                                                            |

| HXC      | Type     | Pri | Status | Title                                                                                               |
|----------|----------|-----|--------|-----------------------------------------------------------------------------------------------------|
|          |          |     |        | powergater/<br>edgeheartbeat CPU<br>temp                                                            |
| HXC-930  | Task     | P2  | Queued | Live discrete-GPU/<br>NPU hardware<br>probe for pkg/<br>resources<br>accelerators                   |
| HXC-1323 | Feature  | P3  | Queued | Implement webOS<br>smart-TV JS agent<br>over WebSocket<br>(DEFERRED<br>device)                      |
| HXC-1326 | Research | P3  | Queued | Implement pkg/<br>quantum Qiskit<br>Runtime plugin for<br>async circuit<br>submission<br>(DEFERRED) |
| HXC-1330 | Docs     | P3  | Queued | Define and validate<br>5 reference cluster<br>build recipes<br>(\$250/\$500/\$1k/<br>\$2k/\$5k)     |
| HXC-1620 | Bug      | P3  | Queued | Harden tierdetect<br>linux<br>cpuinfoIsARM64<br>multi-digit CPU-<br>architecture parsing            |

## Phase 6 (17 active)

| HXC      | Type    | Pri | Status | Title                                                                               |
|----------|---------|-----|--------|-------------------------------------------------------------------------------------|
| HXC-1369 | Feature | P0  | Queued | Implement<br>cmd/helix-<br>federation<br>binary and<br>helixctl<br>federation CLI   |
| HXC-1384 | Task    | P0  | Queued | Establish<br>multi-cell<br>integration<br>testbed for<br>CLAUDE-1<br>usability gate |
| HXC-933  | Feature | P0  | Queued | Build cmd/<br>helix-<br>federation and                                              |

| HXC      | Type    | Pri | Status | Title                                                                                    |
|----------|---------|-----|--------|------------------------------------------------------------------------------------------|
|          |         |     |        | wire Phase 6 packages (federation/crdt/hlc/spiffefed/gitops)                             |
| HXC-1362 | Feature | P1  | Queued | Implement pkg/cilium Cluster Mesh client and global-service propagation                  |
| HXC-1370 | Feature | P1  | Queued | Extend internal/node cell agent: gossip-tier selection + federation identity attestation |
| HXC-1385 | Task    | P1  | Queued | Validate Phase 6 exit-gate KPIs against the live federation                              |
| HXC-1354 | Feature | P2  | Queued | Implement SSH tunnel bridge fallback for control-plane bootstrap                         |
| HXC-1355 | Feature | P2  | Queued | Implement cloud VPN bridge with provider-specific endpoint discovery                     |
| HXC-1360 | Feature | P2  | Queued | Integrate External Secrets Operator + Vault for federation secrets                       |
| HXC-1366 | Feature | P2  | Queued | Implement cloud bursting auto-scaling burst pool with spot                               |

| HXC      | Type    | Pri | Status | Title                                                                            |
|----------|---------|-----|--------|----------------------------------------------------------------------------------|
|          |         |     |        | termination handling                                                             |
| HXC-1378 | Feature | P2  | Queued | Implement hierarchical Prometheus federation across cells                        |
| HXC-1379 | Feature | P2  | Queued | Implement OpenTelemetry cross-cell distributed tracing                           |
| HXC-1382 | Feature | P2  | Queued | Implement Velero tiered disaster recovery backup and restore                     |
| HXC-922  | Task    | P2  | Queued | Live ArgoCD 3-cell ApplicationSet rollout usability capture (HXC-1367 live gate) |
| HXC-924  | Task    | P2  | Queued | Live Karmada multi-cluster <60s failover usability capture (HXC-1364 live gate)  |
| HXC-925  | Task    | P2  | Queued | Two-node LAN mDNS discovery usability capture (HXC-1347 live gate)               |
| HXC-1376 | Docs    | P3  | Queued | Establish quarterly Game Day federation chaos drill protocol                     |

## Phase 7 (4 active)

| HXC      | Type    | Pri | Status | Title                                                                          |
|----------|---------|-----|--------|--------------------------------------------------------------------------------|
| HXC-1414 | Feature | P2  | Queued | Implement NATS leaf-node edge-to-core store-and-forward topology (G-12, P2)    |
| HXC-1619 | Task    | P2  | Queued | Enable cmd/dst-sim 1000-seed gate as a live CI job                             |
| HXC-1423 | Docs    | P3  | Queued | Author TLA+ formal specifications for consensus/coordination protocols (P2)    |
| HXC-914  | Task    | P3  | Queued | internal/chaos: author live-cluster nightly pipeline + GitHub Actions workflow |

## Phase 8 (40 active)

| HXC      | Type    | Pri | Status | Title                                                                       |
|----------|---------|-----|--------|-----------------------------------------------------------------------------|
| HXC-1434 | Feature | P0  | Queued | Implement GraVal-style miner attestation handshake controller in pkg/chutes |
| HXC-1438 | Feature | P0  | Queued | Implement pkg/chutes MinerController K3s deployment lifecycle               |
| HXC-1440 | Feature | P0  | Queued | Deploy PostgreSQL StatefulSet for miner inventory tracking                  |

| HXC      | Type    | Pri | Status | Title                                                                 |
|----------|---------|-----|--------|-----------------------------------------------------------------------|
| HXC-1441 | Feature | P0  | Queued | Deploy Redis pub/sub Deployment for miner event propagation           |
| HXC-1442 | Feature | P0  | Queued | Deploy GraVal bootstrap privileged DaemonSet for GPU attestation      |
| HXC-1443 | Feature | P0  | Queued | Deploy miner-api Deployment with NodePort 32000                       |
| HXC-1444 | Feature | P0  | Queued | Deploy Gepetto strategy engine via ConfigMap-backed Deployment        |
| HXC-1445 | Feature | P0  | Queued | Deploy registry-proxy host-network DaemonSet on port 30500            |
| HXC-1450 | Feature | P0  | Queued | Implement pkg/bittensor wallet (coldkey/hotkey) and TAO balance       |
| HXC-1451 | Feature | P0  | Queued | Implement Bittensor SN64 subnet registration flow                     |
| HXC-1462 | Feature | P0  | Queued | Author helm/helixcluster-chutes base chart (values.yaml)              |
| HXC-1465 | Feature | P0  | Queued | Author scripts/chutes-node-prep.sh bare-metal node preparation        |
| HXC-1466 | Feature | P0  | Queued | Author scripts/chutes-miner-deploy.sh miner deployment + registration |
| HXC-1491 | Task    | P0  | Queued | Validate end-to-end full request flow integration                     |



| HXC      | Type    | Pri | Status | Title                                                                |
|----------|---------|-----|--------|----------------------------------------------------------------------|
|          |         |     |        | (workload->client->E2EE->network->miner->response)                   |
| HXC-1463 | Feature | P1  | Queued | Author chute-deployment.yaml<br>Helm template for inference pods     |
| HXC-1464 | Feature | P1  | Queued | Author values-models.yaml<br>with 8 pre-configured model deployments |
| HXC-1467 | Feature | P1  | Queued | Author scripts/chutes-health-monitor.sh<br>observability + alerts    |
| HXC-1468 | Feature | P1  | Queued | Author scripts/chutes-verify.sh<br>end-to-end verification           |
| HXC-1471 | Feature | P1  | Queued | Deploy vLLM inference engine<br>with PagedAttention defaults         |
| HXC-1472 | Feature | P1  | Queued | Deploy SGLang inference engine<br>with RadixAttention                |
| HXC-1485 | Task    | P1  | Queued | Author HelixQA Challenge: miner<br>deployment and TAO earnings       |
| HXC-1486 | Task    | P1  | Queued | Author HelixQA Challenge: E2EE<br>inference roundtrip                |
| HXC-1487 | Task    | P1  | Queued | Author HelixQA Challenge:<br>multi-marketplace routing decision      |
| HXC-1488 | Task    | P1  | Queued | Author HelixQA Challenge:                                            |

| HXC      | Type     | Pri | Status | Title                                                                                                       |
|----------|----------|-----|--------|-------------------------------------------------------------------------------------------------------------|
|          |          |     |        | GraVal<br>attestation pass/<br>fail                                                                         |
| HXC-1489 | Task     | P1  | Queued | Validate Chutes<br>API client P99<br>first-token<br>streaming<br>latency < 500ms                            |
| HXC-935  | Feature  | P1  | Queued | Wire Phase 8<br>marketplace/<br>GraVal/miner<br>into a real<br>control loop                                 |
| HXC-936  | Bug      | P1  | Queued | Resolve dual<br>package trees:<br>top-level pkg/<br>gputest vs<br>security/pkg/<br>gputest (and<br>similar) |
| HXC-1432 | Feature  | P2  | Queued | Implement gzip<br>compression<br>pipeline in E2EE<br>inference<br>envelope                                  |
| HXC-1433 | Task     | P2  | Queued | Reconcile E2EE<br>AEAD spec<br>deviation<br>(AES-256-GCM<br>vs roadmap<br>ChaCha20-<br>Poly1305)            |
| HXC-1437 | Research | P2  | Queued | Implement<br>GraVal CUDA<br>Proof-of-<br>Consecutive-<br>VRAM-Work<br>kernel<br>(DEFERRED,<br>GPU)          |
| HXC-1452 | Feature  | P2  | Queued | Implement Yuma<br>Consensus /<br>metagraph<br>weight queries in<br>pkg/bittensor                            |
| HXC-1457 | Feature  | P2  | Queued | Implement io.net<br>marketplace<br>adapter                                                                  |
| HXC-1459 | Feature  | P2  | Queued |                                                                                                             |

| HXC      | Type    | Pri | Status | Title                                                                              |
|----------|---------|-----|--------|------------------------------------------------------------------------------------|
|          |         |     |        | Implement Salad marketplace adapter                                                |
| HXC-1475 | Feature | P2  | Queued | Deploy Intel TDX + NVIDIA CC TEE stack via sek8s                                   |
| HXC-1477 | Feature | P2  | Queued | Implement Cosign/Sigstore image admission controller for K3s                       |
| HXC-1490 | Task    | P2  | Queued | Achieve >60% line coverage on pkg/chutes, pkg/marketplace, pkg/e2ee (Codecov gate) |
| HXC-1453 | Feature | P3  | Queued | Implement child-hotkey delegation support in pkg/bittensor                         |
| HXC-1474 | Feature | P3  | Queued | Deploy TurboDiffusion video-generation engine                                      |
| HXC-1478 | Feature | P3  | Queued | Implement LUKS-encrypted root for miner nodes                                      |
| HXC-1479 | Feature | P3  | Queued | Implement Cilium egress control and Watchtower integrity challenges                |

## Phase 8B (28 active)

| HXC      | Type    | Pri | Status | Title                                                            |
|----------|---------|-----|--------|------------------------------------------------------------------|
| HXC-1492 | Feature | P0  | Queued | Define pkg/pool GPUProvider interface as single integration seam |
| HXC-1521 | Task    | P0  | Queued | Expose security ML-KEM-768 E2EE Session/                         |

| HXC      | Type    | Pri | Status | Title                                                                              |
|----------|---------|-----|--------|------------------------------------------------------------------------------------|
|          |         |     |        | Transport as pkg/<br>e2ee                                                          |
| HXC-1522 | Feature | P0  | Queued | Implement pkg/<br>e2ee E2EProxy<br>outbound remote-<br>GPU traffic proxy           |
| HXC-1552 | Task    | P0  | Queued | Implement<br>HelixQA<br>BurstChallenge<br>saturate+failover<br>exit gate           |
| HXC-1553 | Task    | P0  | Queued | Implement E2E<br>helix submit -><br>saturate -> route-<br>to-Chutes demo           |
| HXC-1520 | Feature | P1  | Queued | Implement<br>LocalGPURegistrar<br>nvidia-smi<br>discovery + GPU<br>classification  |
| HXC-1526 | Feature | P1  | Queued | Implement pkg/<br>proxy CUDA API<br>interceptor +<br>memory staging                |
| HXC-1528 | Feature | P1  | Queued | Implement virtual /<br>dev/nvidia* device<br>creation proxying<br>to remote GPU    |
| HXC-1536 | Feature | P1  | Queued | Register virtual<br>GPU devices into<br>internal/node<br>cluster resource<br>model |
| HXC-1537 | Task    | P1  | Queued | Integrate<br>GraValVerifier into<br>pkg/security<br>provider admission             |
| HXC-1540 | Feature | P1  | Queued | Implement local<br>vLLM + SGLang<br>serving stack<br>deployment                    |
| HXC-1541 | Feature | P1  | Queued | Implement GPU-<br>proxy DaemonSet<br>advertising virtual<br>nvidia.com/gpu         |
| HXC-1549 | Task    | P1  | Queued | Author Helm chart<br>for Phase 8B GPU<br>pool + burst<br>deployment                |

| HXC      | Type    | Pri | Status | Title                                                                    |
|----------|---------|-----|--------|--------------------------------------------------------------------------|
| HXC-1554 | Task    | P1  | Queued | Stand up real-provider CI integration (Chutes/io.net/RunPod)             |
| HXC-1555 | Task    | P1  | Queued | Capture E2EE usability evidence: tee=true encrypted-traffic gate         |
| HXC-1557 | Task    | P1  | Queued | Run inference p99 <500ms load test via Chutes (1000 requests)            |
| HXC-1615 | Feature | P1  | Queued | Wire pkg/gpu ProviderAdapter registry into pkg/pool provider consumption |
| HXC-1517 | Feature | P2  | Queued | Implement AWS Spot 2-minute interrupt -> CRIU/S3 checkpoint failover     |
| HXC-1524 | Feature | P2  | Queued | Implement pkg/e2ee TDX AttestationVerifier for sensitive workloads       |
| HXC-1525 | Feature | P2  | Queued | Implement key rotation CronJob + HSM integration for E2EE keys           |
| HXC-1543 | Feature | P2  | Queued | Implement KEDA queue-depth/custom-metric autoscaling                     |
| HXC-1545 | Feature | P2  | Queued | Implement CRIU GPU-context checkpoint/restore + graceful degradation     |
| HXC-1550 | Task    | P2  | Queued | Author Docker Compose development stack for Phase 8B                     |
| HXC-1558 | Task    | P2  | Queued | Run 48-hour load test and 7-day soak (>99.9% success)                    |
| HXC-1559 | Task    | P2  | Queued |                                                                          |

| HXC      | Type    | Pri | Status | Title                                                     |
|----------|---------|-----|--------|-----------------------------------------------------------|
|          |         |     |        | Run chaos suite for provider failover and DDoS resilience |
| HXC-1518 | Docs    | P3  | Queued | Document GCP and Azure provider adapter patterns          |
| HXC-1542 | Feature | P3  | Queued | Implement bittencert blockchain-backed X.509 identity     |
| HXC-1546 | Feature | P3  | Queued | Implement chutes-dropzone self-hosted gateway integration |

## Phase 8C (12 active)

| HXC      | Type    | Pri | Status | Title                                                                             |
|----------|---------|-----|--------|-----------------------------------------------------------------------------------|
| HXC-1560 | Feature | P0  | Queued | Implement pkg/security/e2ee PQ envelope (ML-KEM-768 + HKDF-SHA256)                |
| HXC-1562 | Task    | P0  | Queued | Add Chutes byte-interop test-vector corpus for pkg/security/e2ee                  |
| HXC-1574 | Feature | P0  | Queued | Implement pkg/security/ attestation TEE quote construct/verify                    |
| HXC-1575 | Feature | P0  | Queued | Implement Intel TDX quote generation with nonce+cert binding (report_data layout) |
| HXC-1576 | Feature | P0  | Queued | Implement NVIDIA GPU attestation evidence verification (NRAS/nvTrust)             |
| HXC-1566 | Feature | P1  | Queued | Implement E2EE instance-                                                          |

| HXC      | Type     | Pri | Status | Title                                                                                  |
|----------|----------|-----|--------|----------------------------------------------------------------------------------------|
|          |          |     |        | discovery + single-use nonce contract                                                  |
| HXC-1577 | Feature  | P1  | Queued | Implement measured-boot / RTMR3 access-config measurement                              |
| HXC-1578 | Feature  | P1  | Queued | Implement attestation-gated key release (LUKS/secret) handshake                        |
| HXC-1591 | Feature  | P1  | Queued | Extend pkg/resources GPUInfo with device fingerprint + attested/thermal state          |
| HXC-1594 | Feature  | P1  | Queued | Implement LLMOrchestrator vLLM/SGLang backend WRAP (OpenAI HTTP)                       |
| HXC-1593 | Feature  | P2  | Queued | Implement VRAM-capacity ( $\geq 95\%$ ) usable-memory gate in resources/gpuattest      |
| HXC-1611 | Research | P3  | Queued | Implement Karmada-pattern multi-cluster aggregation over SWIM+Raft+etcd (Phase 6 hook) |